NORDHAVEN

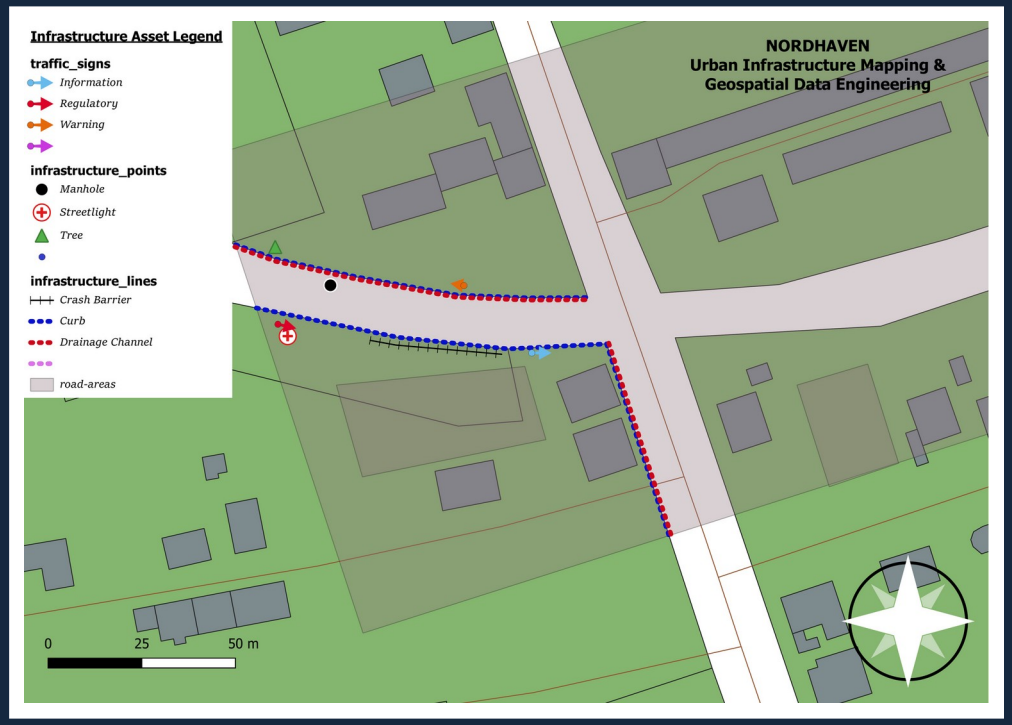
Urban Infrastructure Mapping & Geospatial Data Engineering

Bremen, Germany

Independent Geospatial Portfolio Case Study

Prepared by Mahadev M

QGIS | Spatial Data Engineering | Infrastructure Mapping



Final infrastructure overview map (QGIS print layout)

Independent demonstration project. Not a client deliverable, an official municipal inventory or confidential employer data.

Project Overview & Objectives

What it is

An independently developed infrastructure mapping case study that combines OpenStreetMap-derived reference layers with demonstration infrastructure layers in a structured GeoPackage database.

Why it was built

To demonstrate practical geospatial data production skills end to end, based on professional experience in infrastructure data production.

Main objectives

- Develop a structured geospatial database for infrastructure features
- Organise data with suitable geometry types and attribute schemas
- Demonstrate polygon, point and line feature digitisation
- Apply attribute management and basic spatial quality checks
- Produce a professional cartographic output in QGIS

Spatial data preparation

Infrastructure digitisation

Attribute management

Geospatial database development

Basic spatial QA/QC

Cartographic production

Study area: Bremen, Germany **Platform:** QGIS 3.10.13
CRS: ETRS89 / UTM 32N (EPSG:25832) **Database:** GeoPackage

Data Sources & GIS Architecture

1. **OSM Reference Data**
   Roads, buildings and land-use layers (OSM-derived)
2. **QGIS Project Environment**
   QGIS 3.10.13 · working CRS EPSG:25832
3. **Infrastructure Feature Digitisation**
   Polygon, point and line demonstration features
4. **Attribute Management**
   Identifiers, classes, materials, source, review status
5. **GeoPackage Spatial Database**
   Nordhaven_Infrastructure.gpkg · five stored layers
6. **QA/QC and Cartographic Output**
   Basic validation checks · print layout exported to PDF

1 OSM-derived reference layers

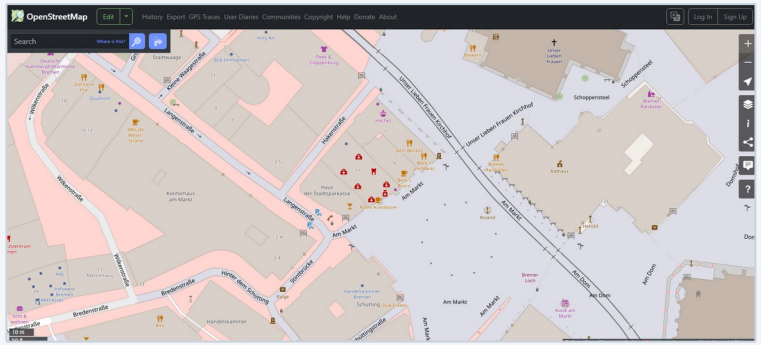
bremen_gis_osm_roads_free · bremen_gis_osm_buildings_a_free · bremen_gis_osm_landuse_a_free

2 Independent demonstration features

road-areas · infrastructure_points · infrastructure_lines · traffic_signs

3 Resulting GIS database and outputs

Nordhaven_Infrastructure.gpkg · QGIS project · infrastructure overview map (PDF)



OpenStreetMap view of Bremen city centre, shown to illustrate OSM reference content. Study area: Bremen, Germany.

© OpenStreetMap contributors | Open Database License (ODbL) 1.0 | openstreetmap.org/copyright

Infrastructure Mapping Workflow

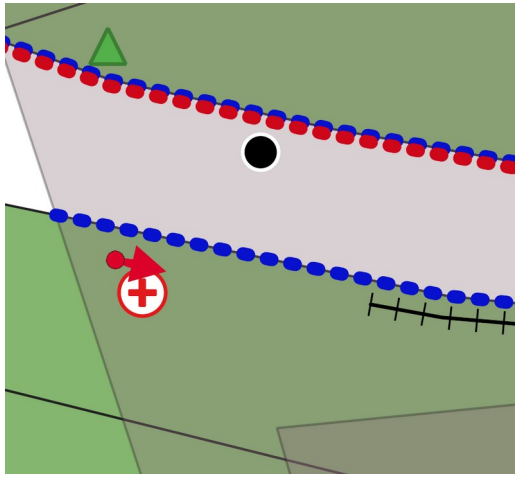


POLYGON GEOMETRY

Road-area polygons

Parking area, pedestrian area and roadway with surface materials.

3 records

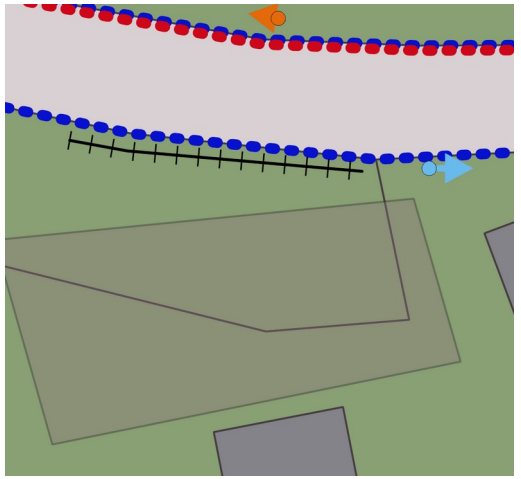


POINT GEOMETRY

Infrastructure points

Manhole, streetlight and tree assets with status and material.

3 records

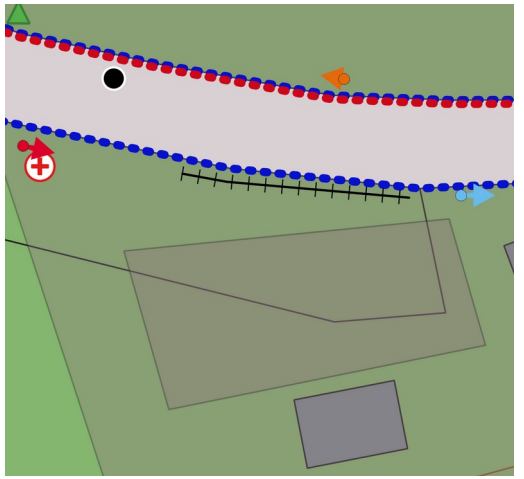


LINE GEOMETRY

Infrastructure lines

Curbs, drainage channels and crash barriers with condition.

6 records



LINE GEOMETRY

Traffic-sign features

Synthetic regulatory, warning and information signs. Stored as line geometry in the current database, not as a point layer.

3 records

Attribute Management & Database Structure

Road areas · 3 records

road-areas :: Features Total: 3, Filtered: 3, Selected: 0

	fid	feature_id	area_class	surface_material	area_m2	source_ref	review_status
1	3	3	Parking Area	Concrete	551.18	OSM_parking_t...	Pending
2	4	4	Pedestrian Area	Concrete	1138.74	OSM_pedestria...	Pending
3	2	2	Roadway	Asphalt	30937.37	OSM_road_test...	Pending

Infrastructure points · 3 records

infrastructure_points :: Features Total: 3, Filtered: 3, Selected: 0

	fid	feature_id	asset_type	asset_status	material	source_ref	review_status	remarks
1	2	2	Manhole	Active	Concrete	OSM_point_test...	Pending	NULL
2	3	3	Tree	Active	Other	OSM_point_test...	Pending	NULL
3	1	1	Streetlight	Active	Metal	OSM_point_test...	Pending	NULL

Infrastructure lines · 6 records

infrastructure_lines :: Features Total: 6, Filtered: 6, Selected: 0

	fid	feature_id	infrastructure_type	material	condition	source_ref	review_status	remarks
1	1	1	Curb	Concrete	Good	manual_digitisation_01	Pending	Sample curb feature
2	2	2	Curb	Concrete	Good	manual_digitisation_02	Pending	Sample curb segment 02
3	3	3	Drainage Channel	Concrete	Good	manual_digitisation_03	Pending	Sample drainage alignment
4	4	4	Curb	Concrete	Good	manual_digitisation_04	Pending	Sample curb segment 03
5	5	5	Drainage Channel	Concrete	Good	manual_digitisation_05	Pending	Sample drainage alignment 02
6	6	6	Crash Barrier	Metal	Good	manual_digitisation_06	Pending	Sample crash barrier segment

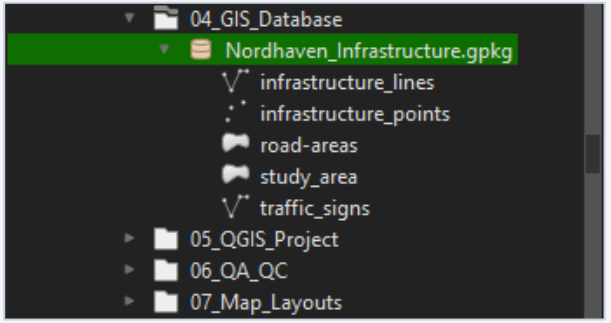
Traffic signs · 3 records

traffic_signs :: Features Total: 3, Filtered: 3, Selected: 0

	fid	feature_id	sign_category	sign_code	sign_text	mounting_type	orientation	source_ref	review_status	remarks
1	1	1	Regulatory	R-01	STOP	Single Pole	Facing road	synthetic_demo_01	Pending	Synthetic demonstration feature
2	2	2	Warning	W-01	Curve Ahead	Single Pole	Facing road	synthetic_demo_02	Pending	Synthetic demonstration feature
3	3	3	Information	I-01	Parking	Single Pole	Facing road	synthetic_demo_03	Pending	Synthetic demonstration feature

Nordhaven_Infrastructure.gpkg

- infrastructure_lines — line
- infrastructure_points — point
- road-areas — polygon
- study_area — polygon
- traffic_signs — line



QGIS Browser panel: the five stored layers.

Stored name is road-areas (hyphen); earlier notes may write road_areas. The stored name is used throughout.

Field	Purpose
feature_id	Unique feature identifier
area_class · asset_type · infrastructure_type · sign_category	Infrastructure classification
surface_material · material · condition	Material and condition description
source_ref	Source reference for each record
review_status	Review state (Pending on all records shown)

Structured attributes make every feature identifiable, classifiable and reviewable. One GeoPackage keeps all layers in a single portable file.

Quality Assurance & Quality Control

Geometry validity

0

invalid geometries in the tested layers

QGIS Check Validity run on selected geometry layers.

Attribute completeness

0

road-area records with missing values

Query on road-areas checked area_class, surface_material, area_m2, source_ref and review_status. It returned zero selected features.

Visual review

Visual

inspection of mapped features

Features reviewed in QGIS for representation and consistency with the OSM reference layers.

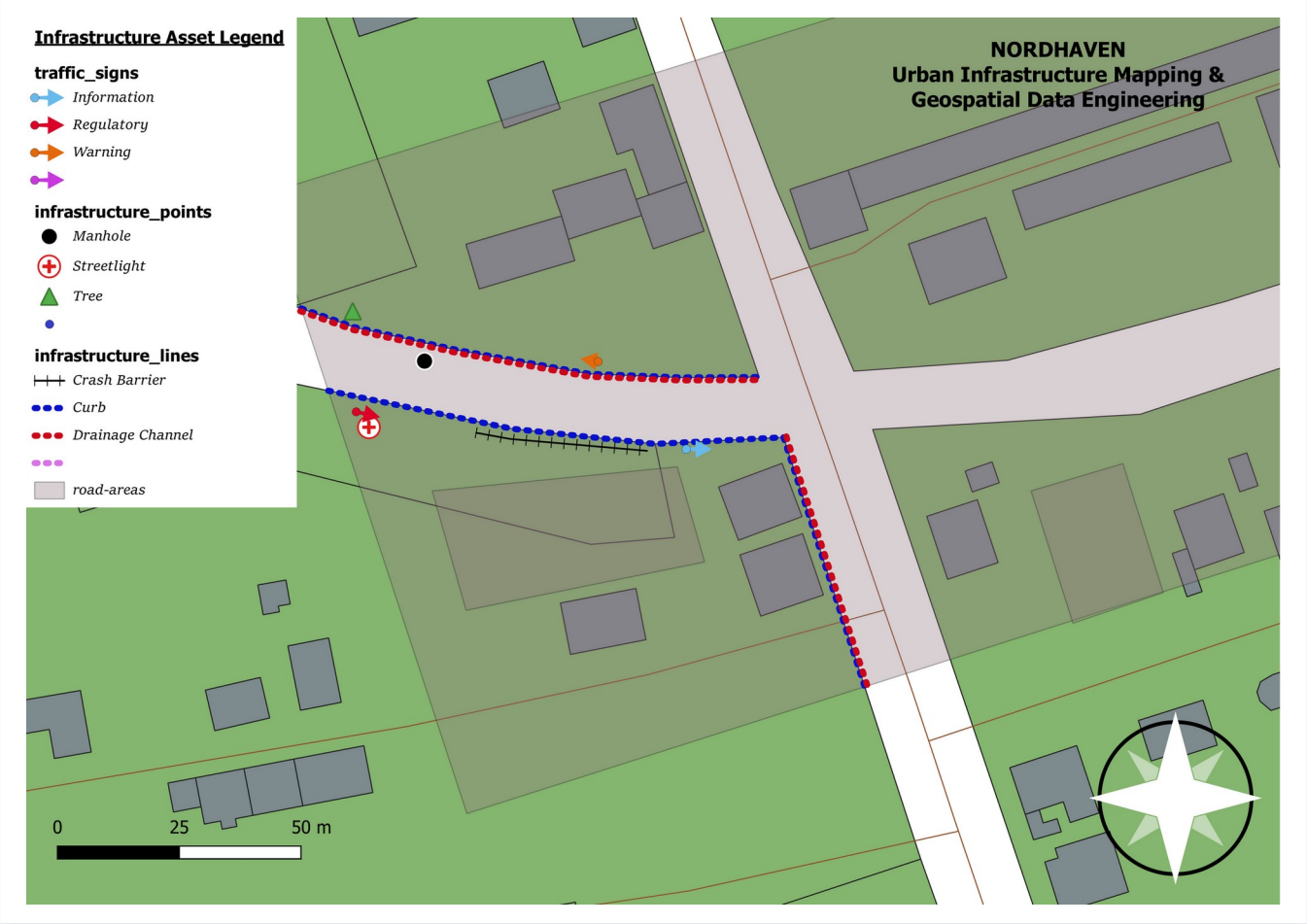
Demonstration-level validation

These are basic checks completed within this project. They are not a complete assessment of positional or thematic accuracy and do not represent production certification.

Not performed in this project

- Positional accuracy assessment
- Topology validation
- Automated Python QA/QC
- PostGIS validation

Final Map & Project Outputs



Infrastructure overview map: QGIS print layout exported to PDF

Asset representation: Point symbols, line styles and sign arrows.

Feature categorisation: Legend grouped by layer and feature type.

Spatial context: OSM buildings, roads and land use as backdrop.

Cartographic presentation: Title, legend, scale bar and north arrow.

Project outputs

- **Nordhaven_Infrastructure.gpkg** GeoPackage
- **Nordhaven_Master.qgz** QGIS project
- **Infrastructure Overview Map** PDF
- **Technical Methodology Document** Documentation

Conclusion & Future Enhancements

Capabilities demonstrated

- Spatial data preparation in QGIS (EPSG:25832)
- Polygon, point and line feature digitisation
- Project-specific attribute schemas
- GeoPackage spatial database organisation
- Basic geometry and attribute QA/QC
- Cartographic production with a PDF map

Proposed future enhancements

Not yet implemented in this project

- Python automation
- GeoPandas geospatial processing
- PostGIS spatial database integration
- Advanced spatial QA/QC
- Automated validation reporting

NORDHAVEN

Geospatial Infrastructure Mapping & Data Engineering

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